

Problem Definition

The Problem

- The Logistics companies in South Africa regularly face delivery delays caused by traffic congestion, vehicle breakdowns, bad weather, port congestion, and warehouse issues. These disruptions often happen at once, and staff currently have no automated way to determine which delay is most urgent.

Relevance to the theme

- This project fits the theme of the Fourth Industrial Revolution because it uses an Artificial Intelligence (AI) solution to solve a real industry problem, using synthetic data rather than real customer information. The problem is highly relevant because it applies AI techniques to a well-documented challenge in the logistics sector.

Benefit of Solving the Problem

- An AI-driven triage system will classify delays automatically into priority levels and present them through a simple dashboard showing delay risks, urgency, location, and recommended action. This would:
 - i. Reduce the manual effort required to assess each delay
 - ii. Improve consistency in how delays are prioritized

This makes the problem not only realistic and achievable within the project scope, but also clearly beneficial in demonstrating measurable business value.